

Employee Attrition Prediction & HR Analytics

Project Report

Prepared By

Shrutee Revankar

1. Project Overview

Employee attrition is one of the biggest challenges faced by organizations as it directly impacts productivity, recruitment costs, employee morale, and business continuity. This project aims to analyze historical employee data, identify the key factors influencing employee attrition, and build a machine learning model capable of predicting whether an employee is likely to leave the organization.

The project combines **Python, Machine Learning, Streamlit, and Power BI** to create a complete HR Analytics solution. Power BI provides interactive visualizations for business insights, while the Streamlit application enables real-time attrition prediction based on employee attributes.

2. Objectives

The main objectives of this project are:

- Analyze employee demographics and organizational data.
 - Identify the major causes of employee attrition.
 - Build an interactive HR analytics dashboard.
 - Develop a machine learning model to predict employee attrition.
 - Help HR managers make proactive decisions to improve employee retention.
-

3. Technology Stack

Component	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Data Visualization	Power BI, Plotly
Machine Learning	Scikit-learn
Model	Random Forest Classifier
Web Application	Streamlit
Dataset	HR Employee Attrition Dataset

4. Project Workflow

The project follows the complete data analytics and machine learning lifecycle:

1. Data Collection
 2. Data Cleaning
 3. Data Preprocessing
 4. Feature Engineering
 5. Exploratory Data Analysis
 6. Power BI Dashboard Development
 7. Machine Learning Model Training
 8. Model Evaluation
 9. Streamlit Application Development
 10. Employee Attrition Prediction
-

5. Data Cleaning & Feature Engineering

The dataset was cleaned by handling missing values, correcting data types, removing duplicates, and standardizing categorical variables.

Additional features were created to improve business understanding:

- Age Group
- Income Category
- Experience Level
- Distance Category
- Overtime Flag
- Attrition Flag
- High Performer
- Long Tenure Employee

These engineered features enhanced both dashboard insights and machine learning performance.

6. Power BI Dashboard Analysis

The Power BI dashboard provides an overview of employee attrition through interactive KPIs and visualizations.

Key KPIs

- **Total Employees:** 1,470
- **Employees Left:** 237
- **Attrition Rate:** 16.12%
- **Average Age:** 36.92 Years
- **Average Monthly Income:** ₹6,502

The dashboard also includes interactive filters for Department, Gender, Job Role, Education Field, Business Travel, and Marital Status.

7. Dashboard Insights

Overall Attrition

Out of **1,470 employees**, **237 employees left** the organization, resulting in an attrition rate of approximately **16%**. While the majority of employees remain with the company, this level of attrition is significant enough to warrant proactive retention strategies.

Department-wise Attrition

The **Research & Development department** has the highest number of employees and also records the highest number of employees leaving. Although this department experiences the largest volume of attrition, it is also the largest workforce within the organization. The **Sales department** shows the next highest attrition, while **Human Resources** records the lowest number of employee exits.

Job Role Analysis

The **Sales Executive** role experiences the highest employee turnover, followed by **Laboratory Technician** and **Research Scientist**. These roles may involve higher workloads, performance pressure, or limited career progression, making them more susceptible to attrition.

Overtime Analysis

Employees who frequently work overtime demonstrate a noticeably higher attrition rate compared to employees who do not work overtime. This suggests that excessive workloads and poor work-life balance are major contributors to employee turnover.

Age Group Analysis

The workforce is primarily concentrated in the **26–35** and **36–45** age groups. These groups also account for the highest number of employees leaving the organization. Employees in these age ranges are often in critical career growth stages and may seek better compensation, promotions, or professional development opportunities.

Income Analysis

Employees categorized in the **Low-Income** segment exhibit the highest attrition compared to Medium and High-income groups. This indicates that salary and compensation play an important role in employee retention.

8. Machine Learning Model

A Random Forest Classifier was developed to predict employee attrition.

The model was trained using historical employee information including:

- Age
- Department
- Job Role
- Business Travel
- Monthly Income
- Overtime
- Work Life Balance
- Years at Company
- Job Satisfaction
- Environment Satisfaction
- Distance from Home

The trained model enables HR professionals to predict whether an employee is likely to leave based on their profile.

9. Streamlit Application

The Streamlit application allows HR users to enter employee details and receive an instant attrition prediction.

Features

- Employee information input form
- Real-time prediction
- Clean user interface
- Interactive navigation
- Model explanation
- Prediction probability
- HR decision support

The application converts complex machine learning outputs into actionable business insights, making predictive analytics accessible to non-technical HR professionals.

10. Business Insights

The analysis reveals several factors associated with higher attrition risk:

- Employees working overtime are more likely to leave.
- Lower-income employees have higher turnover rates.
- Sales-related job roles experience greater attrition.
- Mid-career employees (26–45 years) are the most affected.
- Research & Development and Sales departments require focused retention efforts.

These insights help HR prioritize interventions where they will have the greatest impact.

11. Predictive Analysis

The machine learning model enables early identification of employees who may be at risk of leaving.

Predicted high-risk employees generally share characteristics such as:

- Frequent overtime
- Lower salary levels
- Fewer years with the company
- High business travel frequency
- Lower job satisfaction
- Limited work-life balance

This predictive capability allows HR teams to intervene before employees decide to resign.

12. Recommendations

Based on the analysis, the following actions are recommended:

- Reduce excessive overtime by balancing workloads and optimizing staffing.
 - Review compensation structures, particularly for lower-income employees, to improve competitiveness.
 - Strengthen employee engagement programs within Sales and Research & Development departments.
 - Provide structured career development plans, mentorship, and internal promotion opportunities for employees in the 26–45 age group.
 - Conduct regular employee satisfaction surveys to identify concerns before they lead to resignations.
 - Introduce flexible work arrangements where possible to improve work-life balance.
 - Develop targeted retention initiatives for high-risk employees identified by the predictive model.
 - Implement recognition and reward programs to acknowledge employee contributions and improve morale.
-

13. Future Scope

This project can be extended with additional capabilities, including:

- Real-time HR database integration for continuous monitoring.
- Deployment on cloud platforms such as AWS, Azure, or Google Cloud.
- Automated email alerts for high-risk employee predictions.
- Explainable AI techniques (e.g., SHAP values) to show why the model predicts attrition.
- Advanced predictive models such as XGBoost or LightGBM for improved accuracy.
- HR recommendation engine suggesting personalized retention strategies.
- Department-wise attrition forecasting using time-series analysis.
- Integration with HRMS/ERP systems for automated decision support.
- Executive dashboards with trend analysis and benchmarking.

14. Project Implementation

The implemented solution provides value across multiple organizational levels:

- **HR Managers:** Monitor workforce health and identify retention priorities.
- **Department Heads:** Understand turnover trends within their teams.
- **Leadership:** Use KPI dashboards for strategic workforce planning.
- **Recruitment Teams:** Anticipate hiring needs based on predicted attrition.
- **Employees:** Benefit from improved engagement initiatives, career development, and work-life balance programs.

By combining descriptive analytics with predictive modelling, the system supports data-driven HR decision-making and helps organizations reduce employee turnover while improving workforce stability.

15. Conclusion

The **Employee Attrition Prediction & HR Analytics** project demonstrates an end-to-end analytics solution that integrates data preprocessing, business intelligence, machine learning, and interactive web technologies. The Power BI dashboard provides clear visibility into workforce trends, while the Streamlit application enables real-time attrition prediction. Together, these tools empower HR teams to make informed decisions, proactively address employee retention challenges, and build a more engaged and productive workforce. This project highlights the practical application of data analytics and predictive modeling in solving real-world HR problems.